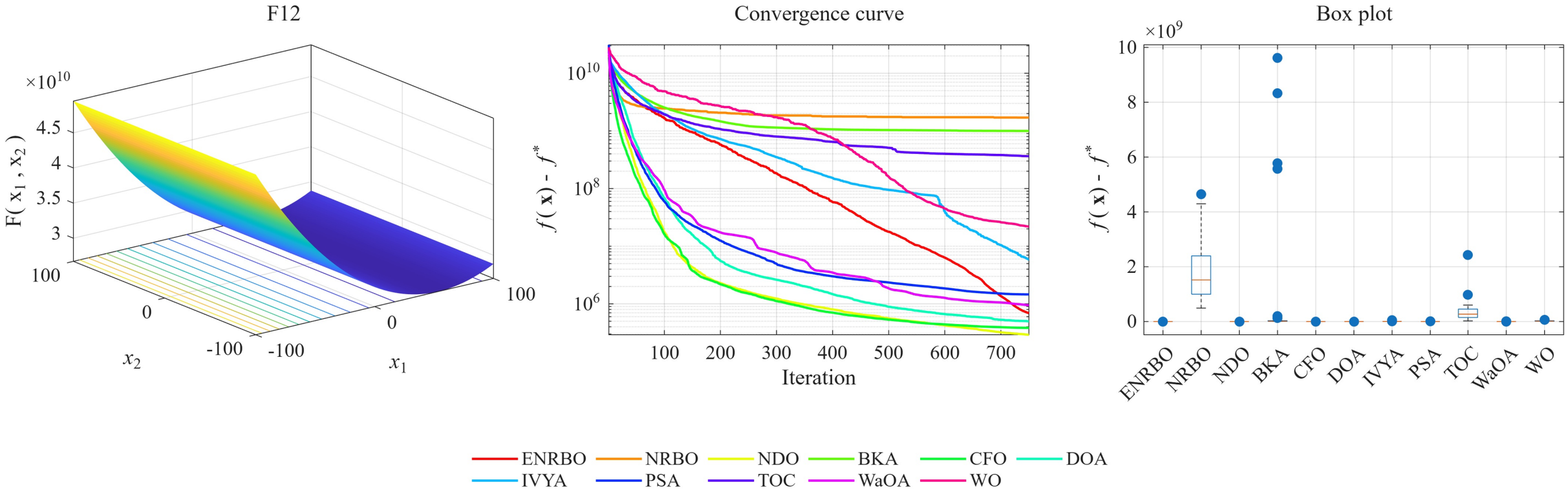

F12
×10^10
4.5
4
3.5
3
F(x1 , x2)
100
0
-100
x2
-100
0
100
x1
Convergence curve
10^10
10^8
10^6
f(x) - f*
100
200
300
400
500
600
700
Iteration
Box plot
×10^9
10
8
6
4
2
0
f(x) - f*
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO
ENRBO
NRBO
NDO
BKA
CFO
DOA
IVYA
PSA
TOC
WaOA
WO